

Supplementary Materials

Stress-Based Validation of Bio-Inspired Phosphene Vision Encoding: Trade-Offs among Performance, Safety, and Representation Stability

This supplementary file contains the supplementary tables cited in the main manuscript.

Supplementary Table S1. Stress operators, sweep levels, and residual safety-violation severity definitions.

Component	Definition	Implementation details
Stress reference	clean	Unperturbed input image, used as the baseline condition
Stress operator	noise	$\mathcal{T}_\alpha^{\text{noise}}(X) = \text{clip}(X + \varepsilon, 0, 1)$, where $\varepsilon \sim \mathcal{N}(0, \sigma(\alpha)^2)$
Noise scale	$\sigma(\alpha)$	$\sigma(\alpha) = 0.6\alpha$
Stress operator	blur	$\mathcal{T}_\alpha^{\text{blur}}(X) = \text{clip}(G_{\sigma(\alpha)} * X, 0, 1)$
Blur scale	$\sigma(\alpha)$	$\sigma(\alpha) = 1.5\alpha$
Stress operator	dropout	$\mathcal{T}_\alpha^{\text{dropout}}(X) = M_\alpha \odot X$, where $M_\alpha(i, j) \sim \text{Bernoulli}(1 - p(\alpha))$
Dropout probability	$p(\alpha)$	$p(\alpha) = 0.6\alpha$
Stress operator	occlusion	A contiguous rectangular patch is zeroed out at a uniformly sampled location
Occlusion patch size	(h_o, w_o)	$(h_o, w_o) = (\max(1, \lfloor 0.6\alpha h \rfloor), \max(1, \lfloor 0.6\alpha w \rfloor))$
Stress-level grid	α	$\{0.00, 0.25, 0.50, 0.75, 1.00\}$
Post-stress stimulus	$U_{\theta, o, \alpha}(X)$	$U_{\theta, o, \alpha}(X) = \Pi_{\text{safe}}(E_\theta(\mathcal{T}_\alpha^o(X)))$
Residual duty violation	sev_{duty}	$\max\left(0, \frac{f \cdot pw - \text{duty}_{\text{max}}}{\text{duty}_{\text{max}}}\right)$
Residual charge violation	$\text{sev}_{\text{charge}}$	$\max\left(0, \frac{\sum A \cdot f \cdot pw - q_{\text{max}}}{q_{\text{max}}}\right)$
Residual sparsity violation	$\text{sev}_{\text{sparse}}$	$\max\left(0, \frac{n_{\text{active}} - k_{\text{max}}}{k_{\text{max}}}\right)$
Scalar severity	sev	$\text{sev} = \max(\text{sev}_{\text{duty}}, \text{sev}_{\text{charge}}, \text{sev}_{\text{sparse}})$
Interpretation	Residual violation	Severity is evaluated after feasibility projection; it therefore denotes residual post-projection violation, not raw pre-projection burden

Supplementary Table S2. Functional and safety metrics used in the structured stress sweep.

Metric group	Symbol / column	Definition	Interpretation
Condition-wise performance	$P(\theta, o, \alpha)$	$\frac{1}{N} \sum_{i=1}^N \mathbb{I}(D_b(Z_{\theta,o,\alpha}(X_i)) = Y_i)$	Classification accuracy at encoder θ , operator o , and level α
Condition-wise safety	$sev_{\max}(\theta, o, \alpha)$	$\max_i sev_i$	Maximum residual severity over the test set
Condition-wise safety	$sev_{\text{mean}}(\theta, o, \alpha)$	$\frac{1}{N} \sum_{i=1}^N sev_i$	Mean residual severity over the test set
Condition-wise safety	$r_{\text{viol}}(\theta, o, \alpha)$	$\frac{1}{N} \sum_{i=1}^N \mathbb{I}(sev_i > 0)$	Fraction of test samples with nonzero residual severity
Clean baseline	$P_{\text{clean}}(\theta)$	$P(\theta, \text{clean}, 0)$	Clean-condition accuracy used as the nominal reference
Clean baseline	$sev_{\max, \text{clean}}(\theta)$	$sev_{\max}(\theta, \text{clean}, 0)$	Maximum residual severity at the clean condition

Metric group	Symbol / column	Definition	Interpretation
Stress-range summary	$\min P(\theta)$	$\min_{o \neq \text{clean}, \alpha} P(\theta, o, \alpha)$	Worst observed classification performance under structured stress
Stress-range summary	$\max \text{sev}_{\max}(\theta)$	$\max_{o \neq \text{clean}, \alpha} \text{sev}_{\max}(\theta, o, \alpha)$	Worst observed residual severity under structured stress
Stress-range summary	$\max r_{\text{viol}}(\theta)$	$\max_{o \neq \text{clean}, \alpha} r_{\text{viol}}(\theta, o, \alpha)$	Largest observed violation rate under structured stress
Export table	TableStress_PerfSafety_ByEncoderOpLevel.csv	Full condition-wise table over encoder, operator, and level	Primary results table from which clean and worst-case summaries are derived
Export table	Table4_Clean_PerfSafety.csv	Clean-condition subset of the full stress table	Used for benchmark-specific clean baselines
Export table	Table5_Worst_PerfSafety.csv	Worst-case summary across structured stress conditions	Used for worst-case benchmark summaries

Supplementary Table S3. Stress-integrated summaries and tri-objective utility used in the released implementation.

Quantity	Symbol	Definition	Notes
Grid spacing	$\Delta\alpha$	0.25	Derived from the implemented stress grid
Operator-averaged accuracy at level α	$\bar{P}(\theta, \alpha)$	Mean of $P(\theta, o, \alpha)$ across non-clean operators o	clean is excluded from the integrated summary
Operator-averaged representation stability at level α	$\bar{R}(\theta, \alpha)$	Mean of $R(\theta, o, \alpha)$ across non-clean operators o	R is stored in the implementation under the TSI column at the condition level
Operator-averaged residual severity at level α	$\overline{\text{sev}}_{\max}(\theta, \alpha)$	Mean of $\text{sev}_{\max}(\theta, o, \alpha)$ across non-clean operators o	Aggregated before stress integration
Stress-integrated performance	$\text{TSI}_P(\theta)$	$\sum_{\alpha} \bar{P}(\theta, \alpha) \Delta\alpha$	Operator-balanced integral of stressed performance
Stress-integrated representation stability	$\text{TSI}_R(\theta)$	$\sum_{\alpha} \bar{R}(\theta, \alpha) \Delta\alpha$	Operator-balanced integral of topology-based stability
Integrated residual safety burden	$\mathcal{R}_{\text{safe}}(\theta)$	$\sum_{\alpha} \overline{\text{sev}}_{\max}(\theta, \alpha) \Delta\alpha$	Integrated worst-case-oriented residual safety burden
Utility weight	λ_P	1	Equal-weight setting in the released

Quantity	Symbol	Definition	Notes
Utility weight	λ_R	1	implementation Equal-weight setting in the released implementation
Utility weight	λ_S	1	Equal-weight setting in the released implementation
Tri-objective utility	$\mathcal{U}(\theta)$	$\lambda_P \text{TSI}_P(\theta) + \lambda_R \text{TSI}_R(\theta) - \lambda_S \mathcal{R}_{\text{safe}}(\theta)$	Final scalar summary used for encoder comparison
Additional worst-case summaries	$\min P, \max \text{sev}_{\max}, \min R$	Recorded per encoder in the tri-objective summary table	Complements the integrated metrics
Export table	Table10_TriObjective_Summary.csv	Per-encoder benchmark-specific tri-objective summary	Used in Section 5.5

Supplementary Table S4. Benchmark-specific topology metrics on EMNIST Letters.

encoder	op	level	dist h0	dist h1	topo dist	TSI	tau scale
rate	clean	0	0	0	0	1	6.7319
rate	clean	0.25	0	0	0	1	6.7319
rate	clean	0.5	0	0	0	1	6.7319
rate	clean	0.75	0	0	0	1	6.7319
rate	clean	1	0	0	0	1	6.7319
rate	noise	0	0	0	0	1	6.7319
rate	noise	0.25	4.6996	1.546	6.2456	0.3954	6.7319

encoder	op	level	dist h0	dist h1	topo dist	TSI	tau scale
rate	noise	0.5	7.469	2.0717	9.5407	0.2424	6.7319
rate	noise	0.75	9.2739	2.4411	11.715	0.1755	6.7319
rate	noise	1	9.0575	2.8648	11.9223	0.1702	6.7319
rate	blur	0	0	0	0	1	6.7319
rate	blur	0.25	0.3322	0.4411	0.7732	0.8915	6.7319
rate	blur	0.5	0.636	0.8224	1.4584	0.8052	6.7319
rate	blur	0.75	1.1326	1.168	2.3006	0.7105	6.7319
rate	blur	1	1.7471	1.1072	2.8544	0.6544	6.7319
rate	dropout	0	0	0	0	1	6.7319
rate	dropout	0.25	1.5943	1.2737	2.868	0.6531	6.7319
rate	dropout	0.5	4.2646	1.8421	6.1067	0.4037	6.7319
rate	dropout	0.75	5.8643	2.6091	8.4734	0.284	6.7319
rate	dropout	1	5.002	2.2163	7.2183	0.3422	6.7319
rate	occlusion	0	0	0	0	1	6.7319
rate	occlusion	0.25	1.5344	1.0033	2.5377	0.6859	6.7319
rate	occlusion	0.5	11.4823	1.7924	13.2747	0.1392	6.7319
rate	occlusion	0.75	11.5822	3.7159	15.2981	0.1031	6.7319
rate	occlusion	1	24.1256	4.7687	28.8943	0.0137	6.7319
sparse	clean	0	0	0	0	1	6.1142
sparse	clean	0.25	0	0	0	1	6.1142
sparse	clean	0.5	0	0	0	1	6.1142
sparse	clean	0.75	0	0	0	1	6.1142
sparse	clean	1	0	0	0	1	6.1142
sparse	noise	0	0	0	0	1	6.1142
sparse	noise	0.25	9.6348	1.9175	11.5523	0.1512	6.1142
sparse	noise	0.5	13.2006	2.1647	15.3653	0.081	6.1142

encoder	op	level	dist h0	dist h1	topo dist	TSI	tau scale
sparse	noise	0.75	13.2006	2.1647	15.3653	0.081	6.1142
sparse	noise	1	13.2006	2.1647	15.3653	0.081	6.1142
sparse	blur	0	0	0	0	1	6.1142
sparse	blur	0.25	0.5119	0.6524	1.1643	0.8266	6.1142
sparse	blur	0.5	1.5366	0.9538	2.4904	0.6654	6.1142
sparse	blur	0.75	2.6676	1.1601	3.8277	0.5347	6.1142
sparse	blur	1	4.4088	1.467	5.8757	0.3825	6.1142
sparse	dropout	0	0	0	0	1	6.1142
sparse	dropout	0.25	4.0811	0.944	5.0251	0.4396	6.1142
sparse	dropout	0.5	2.0965	1.4623	3.5588	0.5587	6.1142
sparse	dropout	0.75	4.5155	1.8371	6.3526	0.3538	6.1142
sparse	dropout	1	4.3229	2.871	7.1939	0.3083	6.1142
sparse	occlusion	0	0	0	0	1	6.1142
sparse	occlusion	0.25	1.4821	1.0512	2.5333	0.6608	6.1142
sparse	occlusion	0.5	3.216	2.2664	5.4824	0.4079	6.1142
sparse	occlusion	0.75	10.3581	4.4599	14.8179	0.0886	6.1142
sparse	occlusion	1	23.4343	4.5139	27.9482	0.0103	6.1142
temporal	clean	0	0	0	0	1	5.9742
temporal	clean	0.25	0	0	0	1	5.9742
temporal	clean	0.5	0	0	0	1	5.9742
temporal	clean	0.75	0	0	0	1	5.9742
temporal	clean	1	0	0	0	1	5.9742
temporal	noise	0	0	0	0	1	5.9742
temporal	noise	0.25	2.4002	1.9101	4.3103	0.486	5.9742
temporal	noise	0.5	5.8257	1.8378	7.6634	0.2773	5.9742
temporal	noise	0.75	7.2088	2.3822	9.591	0.2008	5.9742

encoder	op	level	dist h0	dist h1	topo dist	TSI	tau scale
temporal	noise	1	12.7054	2.4989	15.2043	0.0785	5.9742
temporal	blur	0	0	0	0	1	5.9742
temporal	blur	0.25	0.6594	0.5109	1.1703	0.8221	5.9742
temporal	blur	0.5	0.8977	1.0424	1.9401	0.7227	5.9742
temporal	blur	0.75	1.042	1.1384	2.1804	0.6942	5.9742
temporal	blur	1	1.0244	1.0031	2.0275	0.7122	5.9742
temporal	dropout	0	0	0	0	1	5.9742
temporal	dropout	0.25	1.739	1.6397	3.3787	0.5681	5.9742
temporal	dropout	0.5	4.2659	2.5601	6.8261	0.319	5.9742
temporal	dropout	0.75	4.8962	2.1975	7.0936	0.305	5.9742
temporal	dropout	1	9.8773	2.1875	12.0647	0.1327	5.9742
temporal	occlusion	0	0	0	0	1	5.9742
temporal	occlusion	0.25	1.8597	1.2119	3.0716	0.598	5.9742
temporal	occlusion	0.5	3.3663	1.7561	5.1224	0.4243	5.9742
temporal	occlusion	0.75	8.931	2.9043	11.8352	0.1379	5.9742
temporal	occlusion	1	10.6338	3.8115	14.4453	0.0891	5.9742
optim	clean	0	0	0	0	1	6.5219
optim	clean	0.25	0	0	0	1	6.5219
optim	clean	0.5	0	0	0	1	6.5219
optim	clean	0.75	0	0	0	1	6.5219
optim	clean	1	0	0	0	1	6.5219
optim	noise	0	0	0	0	1	6.5219
optim	noise	0.25	5.7205	1.5618	7.2823	0.3274	6.5219
optim	noise	0.5	8.3515	2.1099	10.4614	0.2011	6.5219
optim	noise	0.75	11.3902	2.6521	14.0422	0.1161	6.5219
optim	noise	1	11.9281	3.0207	14.9488	0.1011	6.5219

encoder	op	level	dist h0	dist h1	topo dist	TSI	tau scale
optim	blur	0	0	0	0	1	6.5219
optim	blur	0.25	0.2833	0.4347	0.7181	0.8957	6.5219
optim	blur	0.5	0.497	1.0145	1.5115	0.7931	6.5219
optim	blur	0.75	0.881	1.2365	2.1175	0.7228	6.5219
optim	blur	1	1.4554	1.1376	2.593	0.6719	6.5219
optim	dropout	0	0	0	0	1	6.5219
optim	dropout	0.25	2.6439	1.8929	4.5367	0.4988	6.5219
optim	dropout	0.5	3.0403	1.4037	4.4441	0.5059	6.5219
optim	dropout	0.75	4.602	2.5867	7.1887	0.3321	6.5219
optim	dropout	1	19.8608	2.4044	22.2651	0.0329	6.5219
optim	occlusion	0	0	0	0	1	6.5219
optim	occlusion	0.25	1.2753	1.1765	2.4519	0.6866	6.5219
optim	occlusion	0.5	3.6054	2.2497	5.8551	0.4075	6.5219
optim	occlusion	0.75	12.053	2.6644	14.7174	0.1047	6.5219
optim	occlusion	1	22.9608	5.4601	28.4209	0.0128	6.5219

Supplementary Table S5. Benchmark-specific topology metrics on the COCO-derived benchmark.

encoder	op	level	dist h0	dist h1	topo dist	TSI	tau scale
rate	clean	0	0	0	0	1	0.1464
rate	clean	0.25	0	0	0	1	0.1464
rate	clean	0.5	0	0	0	1	0.1464
rate	clean	0.75	0	0	0	1	0.1464
rate	clean	1	0	0	0	1	0.1464
rate	noise	0	0	0	0	1	0.1464
rate	noise	0.25	0.105	0.0662	0.1712	0.3107	0.1464
rate	noise	0.5	0.1855	0.0886	0.2741	0.1538	0.1464

encoder	op	level	dist h0	dist h1	topo dist	TSI	tau scale
rate	noise	0.75	0.237	0.0886	0.3255	0.1083	0.1464
rate	noise	1	0.2624	0.0886	0.3509	0.091	0.1464
rate	blur	0	0	0	0	1	0.1464
rate	blur	0.25	0.0266	0.0185	0.0451	0.7351	0.1464
rate	blur	0.5	0.0749	0.0323	0.1072	0.481	0.1464
rate	blur	0.75	0.0528	0.0386	0.0914	0.5356	0.1464
rate	blur	1	0.0588	0.0409	0.0998	0.506	0.1464
rate	dropout	0	0	0	0	1	0.1464
rate	dropout	0.25	0.0773	0.0536	0.131	0.4088	0.1464
rate	dropout	0.5	0.1024	0.0595	0.1619	0.331	0.1464
rate	dropout	0.75	0.0598	0.0381	0.0979	0.5126	0.1464
rate	dropout	1	0.358	0.0656	0.4236	0.0554	0.1464
rate	occlusion	0	0	0	0	1	0.1464
rate	occlusion	0.25	0.0445	0.0378	0.0824	0.5699	0.1464
rate	occlusion	0.5	0.0533	0.0447	0.098	0.5121	0.1464
rate	occlusion	0.75	0.5376	0.0643	0.6019	0.0164	0.1464
rate	occlusion	1	0.3531	0.0784	0.4315	0.0525	0.1464
sparse	clean	0	0	0	0	1	0.1534
sparse	clean	0.25	0	0	0	1	0.1534
sparse	clean	0.5	0	0	0	1	0.1534
sparse	clean	0.75	0	0	0	1	0.1534
sparse	clean	1	0	0	0	1	0.1534
sparse	noise	0	0	0	0	1	0.1534
sparse	noise	0.25	0.1888	0.0328	0.2216	0.2358	0.1534
sparse	noise	0.5	0.2125	0.0328	0.2453	0.202	0.1534
sparse	noise	0.75	0.2517	0.0328	0.2845	0.1565	0.1534

encoder	op	level	dist h0	dist h1	topo dist	TSI	tau scale
sparse	noise	1	0.2517	0.0328	0.2845	0.1565	0.1534
sparse	blur	0	0	0	0	1	0.1534
sparse	blur	0.25	0.0406	0.0191	0.0597	0.6776	0.1534
sparse	blur	0.5	0.05	0.0268	0.0768	0.6061	0.1534
sparse	blur	0.75	0.0457	0.0382	0.0838	0.5789	0.1534
sparse	blur	1	0.0424	0.0274	0.0698	0.6345	0.1534
sparse	dropout	0	0	0	0	1	0.1534
sparse	dropout	0.25	0.0527	0.0256	0.0783	0.6003	0.1534
sparse	dropout	0.5	0.0841	0.0272	0.1112	0.4842	0.1534
sparse	dropout	0.75	0.0645	0.0318	0.0963	0.5337	0.1534
sparse	dropout	1	0.4968	0.0364	0.5332	0.0309	0.1534
sparse	occlusion	0	0	0	0	1	0.1534
sparse	occlusion	0.25	0.166	0.0295	0.1955	0.2795	0.1534
sparse	occlusion	0.5	0.0578	0.0325	0.0903	0.5552	0.1534
sparse	occlusion	0.75	0.2381	0.0214	0.2596	0.1841	0.1534
sparse	occlusion	1	0.1732	0.0727	0.2459	0.2012	0.1534
temporal	clean	0	0	0	0	1	0.1719
temporal	clean	0.25	0	0	0	1	0.1719
temporal	clean	0.5	0	0	0	1	0.1719
temporal	clean	0.75	0	0	0	1	0.1719
temporal	clean	1	0	0	0	1	0.1719
temporal	noise	0	0	0	0	1	0.1719
temporal	noise	0.25	0.1524	0.0875	0.2399	0.2477	0.1719
temporal	noise	0.5	0.1206	0.0957	0.2163	0.2842	0.1719
temporal	noise	0.75	0.2557	0.0957	0.3514	0.1295	0.1719
temporal	noise	1	0.3645	0.0957	0.4602	0.0688	0.1719

encoder	op	level	dist h0	dist h1	topo dist	TSI	tau scale
temporal	blur	0	0	0	0	1	0.1719
temporal	blur	0.25	0.0243	0.0279	0.0522	0.738	0.1719
temporal	blur	0.5	0.0419	0.0588	0.1007	0.5567	0.1719
temporal	blur	0.75	0.0606	0.0472	0.1078	0.5341	0.1719
temporal	blur	1	0.0887	0.0567	0.1454	0.4291	0.1719
temporal	dropout	0	0	0	0	1	0.1719
temporal	dropout	0.25	0.0711	0.0957	0.1668	0.379	0.1719
temporal	dropout	0.5	0.1098	0.0957	0.2055	0.3026	0.1719
temporal	dropout	0.75	0.0626	0.0875	0.1501	0.4177	0.1719
temporal	dropout	1	0.0974	0.0957	0.1931	0.3252	0.1719
temporal	occlusion	0	0	0	0	1	0.1719
temporal	occlusion	0.25	0.0617	0.0862	0.1479	0.4229	0.1719
temporal	occlusion	0.5	0.1573	0.0864	0.2437	0.2423	0.1719
temporal	occlusion	0.75	0.0686	0.0862	0.1548	0.4064	0.1719
temporal	occlusion	1	0.0915	0.0855	0.177	0.3571	0.1719
optim	clean	0	0	0	0	1	0.2025
optim	clean	0.25	0	0	0	1	0.2025
optim	clean	0.5	0	0	0	1	0.2025
optim	clean	0.75	0	0	0	1	0.2025
optim	clean	1	0	0	0	1	0.2025
optim	noise	0	0	0	0	1	0.2025
optim	noise	0.25	0.1342	0.0665	0.2006	0.3712	0.2025
optim	noise	0.5	0.2746	0.0665	0.341	0.1855	0.2025
optim	noise	0.75	0.2993	0.0793	0.3785	0.1542	0.2025
optim	noise	1	0.2859	0.0855	0.3714	0.1597	0.2025
optim	blur	0	0	0	0	1	0.2025

encoder	op	level	dist h0	dist h1	topo dist	TSI	tau scale
optim	blur	0.25	0.027	0.0246	0.0515	0.7752	0.2025
optim	blur	0.5	0.0627	0.0578	0.1205	0.5514	0.2025
optim	blur	0.75	0.0547	0.0462	0.1009	0.6075	0.2025
optim	blur	1	0.0745	0.0463	0.1207	0.5508	0.2025
optim	dropout	0	0	0	0	1	0.2025
optim	dropout	0.25	0.0785	0.062	0.1404	0.4998	0.2025
optim	dropout	0.5	0.0732	0.0665	0.1397	0.5016	0.2025
optim	dropout	0.75	0.2383	0.0665	0.3048	0.2219	0.2025
optim	dropout	1	0.4856	0.0874	0.573	0.059	0.2025
optim	occlusion	0	0	0	0	1	0.2025
optim	occlusion	0.25	0.1084	0.0451	0.1535	0.4685	0.2025
optim	occlusion	0.5	0.1566	0.0476	0.2043	0.3646	0.2025
optim	occlusion	0.75	0.1925	0.0534	0.2459	0.2968	0.2025
optim	occlusion	1	0.1497	0.079	0.2287	0.3232	0.2025

Supplementary Table S6. Fuzzing coverage metrics on EMNIST Letters.

encoder	iter	coverage size	new bins
rate	0	2	2
rate	1	2	0
rate	2	2	0
rate	3	2	0
rate	4	3	1
rate	5	3	0
rate	6	3	0
rate	7	3	0
rate	8	3	0

encoder	iter	coverage size	new bins
rate	9	3	0
rate	10	3	0
rate	11	3	0
rate	12	3	0
rate	13	3	0
rate	14	3	0
rate	15	3	0
rate	16	3	0
rate	17	3	0
rate	18	3	0
rate	19	3	0
rate	20	3	0
rate	21	3	0
rate	22	3	0
rate	23	3	0
rate	24	3	0
rate	25	3	0
rate	26	3	0
rate	27	3	0
rate	28	3	0
rate	29	3	0
rate	30	3	0
rate	31	3	0
rate	32	3	0
rate	33	3	0

encoder	iter	coverage size	new bins
rate	34	3	0
rate	35	4	1
rate	36	5	1
rate	37	5	0
rate	38	5	0
rate	39	5	0
rate	40	5	0
rate	41	5	0
rate	42	5	0
rate	43	5	0
rate	44	5	0
rate	45	5	0
rate	46	5	0
rate	47	5	0
rate	48	5	0
rate	49	5	0
rate	50	5	0
rate	51	5	0
rate	52	5	0
rate	53	5	0
rate	54	5	0
rate	55	5	0
rate	56	5	0
rate	57	5	0
rate	58	5	0

encoder	iter	coverage size	new bins
rate	59	5	0
rate	60	5	0
rate	61	5	0
rate	62	5	0
rate	63	5	0
rate	64	5	0
rate	65	5	0
rate	66	5	0
rate	67	5	0
rate	68	5	0
rate	69	5	0
rate	70	5	0
rate	71	5	0
rate	72	5	0
rate	73	5	0
rate	74	5	0
rate	75	5	0
rate	76	5	0
rate	77	5	0
rate	78	5	0
rate	79	5	0
sparse	0	7	2
sparse	1	7	0
sparse	2	7	0
sparse	3	7	0

encoder	iter	coverage size	new bins
sparse	4	7	0
sparse	5	7	0
sparse	6	7	0
sparse	7	7	0
sparse	8	7	0
sparse	9	7	0
sparse	10	7	0
sparse	11	7	0
sparse	12	7	0
sparse	13	8	1
sparse	14	8	0
sparse	15	8	0
sparse	16	8	0
sparse	17	8	0
sparse	18	8	0
sparse	19	8	0
sparse	20	9	1
sparse	21	9	0
sparse	22	9	0
sparse	23	9	0
sparse	24	9	0
sparse	25	9	0
sparse	26	9	0
sparse	27	9	0
sparse	28	9	0

encoder	iter	coverage size	new bins
sparse	29	9	0
sparse	30	9	0
sparse	31	9	0
sparse	32	9	0
sparse	33	9	0
sparse	34	9	0
sparse	35	9	0
sparse	36	9	0
sparse	37	9	0
sparse	38	9	0
sparse	39	9	0
sparse	40	9	0
sparse	41	9	0
sparse	42	9	0
sparse	43	10	1
sparse	44	10	0
sparse	45	10	0
sparse	46	10	0
sparse	47	10	0
sparse	48	10	0
sparse	49	10	0
sparse	50	10	0
sparse	51	10	0
sparse	52	10	0
sparse	53	10	0

encoder	iter	coverage size	new bins
sparse	54	10	0
sparse	55	10	0
sparse	56	10	0
sparse	57	10	0
sparse	58	10	0
sparse	59	10	0
sparse	60	10	0
sparse	61	11	1
sparse	62	11	0
sparse	63	11	0
sparse	64	11	0
sparse	65	11	0
sparse	66	11	0
sparse	67	11	0
sparse	68	11	0
sparse	69	11	0
sparse	70	11	0
sparse	71	11	0
sparse	72	11	0
sparse	73	11	0
sparse	74	11	0
sparse	75	11	0
sparse	76	11	0
sparse	77	11	0
sparse	78	11	0

encoder	iter	coverage size	new bins
sparse	79	11	0
temporal	0	13	2
temporal	1	13	0
temporal	2	14	1
temporal	3	15	1
temporal	4	15	0
temporal	5	15	0
temporal	6	17	2
temporal	7	17	0
temporal	8	17	0
temporal	9	17	0
temporal	10	17	0
temporal	11	17	0
temporal	12	17	0
temporal	13	17	0
temporal	14	17	0
temporal	15	17	0
temporal	16	17	0
temporal	17	17	0
temporal	18	17	0
temporal	19	17	0
temporal	20	17	0
temporal	21	17	0
temporal	22	17	0
temporal	23	17	0

encoder	iter	coverage size	new bins
temporal	24	17	0
temporal	25	17	0
temporal	26	17	0
temporal	27	17	0
temporal	28	17	0
temporal	29	17	0
temporal	30	17	0
temporal	31	17	0
temporal	32	17	0
temporal	33	17	0
temporal	34	17	0
temporal	35	17	0
temporal	36	17	0
temporal	37	17	0
temporal	38	17	0
temporal	39	17	0
temporal	40	17	0
temporal	41	17	0
temporal	42	17	0
temporal	43	17	0
temporal	44	17	0
temporal	45	17	0
temporal	46	17	0
temporal	47	17	0
temporal	48	17	0

encoder	iter	coverage size	new bins
temporal	49	17	0
temporal	50	17	0
temporal	51	17	0
temporal	52	17	0
temporal	53	17	0
temporal	54	17	0
temporal	55	17	0
temporal	56	17	0
temporal	57	17	0
temporal	58	17	0
temporal	59	17	0
temporal	60	17	0
temporal	61	18	1
temporal	62	18	0
temporal	63	18	0
temporal	64	18	0
temporal	65	18	0
temporal	66	18	0
temporal	67	18	0
temporal	68	18	0
temporal	69	18	0
temporal	70	18	0
temporal	71	18	0
temporal	72	18	0
temporal	73	18	0

encoder	iter	coverage size	new bins
temporal	74	18	0
temporal	75	18	0
temporal	76	18	0
temporal	77	18	0
temporal	78	18	0
temporal	79	18	0
optim	0	20	2
optim	1	20	0
optim	2	20	0
optim	3	20	0
optim	4	20	0
optim	5	20	0
optim	6	20	0
optim	7	20	0
optim	8	20	0
optim	9	20	0
optim	10	20	0
optim	11	20	0
optim	12	20	0
optim	13	20	0
optim	14	20	0
optim	15	20	0
optim	16	20	0
optim	17	20	0
optim	18	20	0

encoder	iter	coverage size	new bins
optim	19	20	0
optim	20	20	0
optim	21	20	0
optim	22	20	0
optim	23	20	0
optim	24	20	0
optim	25	20	0
optim	26	21	1
optim	27	21	0
optim	28	21	0
optim	29	21	0
optim	30	21	0
optim	31	21	0
optim	32	21	0
optim	33	21	0
optim	34	21	0
optim	35	21	0
optim	36	21	0
optim	37	21	0
optim	38	21	0
optim	39	21	0
optim	40	21	0
optim	41	21	0
optim	42	21	0
optim	43	21	0

encoder	iter	coverage size	new bins
optim	44	21	0
optim	45	21	0
optim	46	21	0
optim	47	21	0
optim	48	21	0
optim	49	21	0
optim	50	21	0
optim	51	21	0
optim	52	21	0
optim	53	21	0
optim	54	21	0
optim	55	21	0
optim	56	22	1
optim	57	22	0
optim	58	22	0
optim	59	22	0
optim	60	22	0
optim	61	22	0
optim	62	22	0
optim	63	22	0
optim	64	22	0
optim	65	22	0
optim	66	22	0
optim	67	22	0
optim	68	22	0

encoder	iter	coverage size	new bins
optim	69	22	0
optim	70	22	0
optim	71	22	0
optim	72	22	0
optim	73	22	0
optim	74	22	0
optim	75	22	0
optim	76	22	0
optim	77	22	0
optim	78	22	0
optim	79	22	0

Supplementary Table S7. Top-ranked fuzzing counterexamples on EMNIST Letters.

encoder	iter	seed_idx	op	level	y_true	y_pred	correct	severity	rep dist	score
rate	56	126	occlusion	0.9171	8	11	0	0	79.5903	20.8976
optim	67	43	occlusion	0.9388	11	2	0	3.97e-08	78.3439	20.586
optim	15	44	occlusion	0.9724	17	19	0	0	74.4419	19.6105
temporal	70	58	occlusion	0.9619	15	25	0	5.4	49.4839	18.771
temporal	22	25	occlusion	0.9694	24	19	0	0	70.2793	18.5698
sparse	14	126	occlusion	0.9279	8	11	0	3.97e-08	69.7287	18.4322
rate	54	98	occlusion	0.8017	11	2	0	1.02e-07	68.0644	18.0161
sparse	38	111	occlusion	0.8951	19	9	0	0	65.7803	17.4451
temporal	30	90	occlusion	0.9982	21	19	0	5.4	43.8409	17.3602
sparse	77	25	occlusion	0.9764	24	19	0	3.97e-08	64.9921	17.248
rate	20	44	occlusion	0.9767	17	19	0	0	64.6128	17.1532
temporal	0	44	occlusion	0.8848	17	11	0	0	64.3679	17.092

encoder	iter	seed_idx	op	level	y_true	y_pred	correct	severity	rep dist	score
temporal	23	55	occlusion	0.9782	23	11	0	0.05	63.5188	16.9297
optim	46	44	occlusion	0.7152	17	19	0	0	63.0425	16.7606
rate	48	53	occlusion	0.9014	11	11	1	3.97e-08	66.7519	16.688
optim	72	98	occlusion	0.8327	11	2	0	0	61.6592	16.4148
rate	16	19	occlusion	0.9167	21	15	0	1.64e-07	61.6185	16.4046
sparse	55	94	occlusion	0.9576	8	8	1	0	65.5318	16.3829
temporal	57	90	occlusion	0.8383	21	19	0	0.025	61.3045	16.3511
sparse	75	43	occlusion	0.8736	11	2	0	0	60.7768	16.1942
temporal	76	32	occlusion	0.8896	5	19	0	0	60.6994	16.1748
temporal	21	32	occlusion	0.8391	5	5	1	1.02e-07	63.4996	15.8749
temporal	6	11	occlusion	0.9589	15	11	0	5.4	37.8928	15.8732
temporal	7	102	occlusion	0.9935	5	25	0	5.4	37.2022	15.7005